Genset ECU — Reconciled Specification

Date: 2026-09-13 **Status:** Phase 0 — supersedes all four prior artifacts
Target: Functional replacement for the `ENGINE CONTROL UNIT` block of Kirloskar wiring diagram `08-GP3-60-001`, on a 25 kVA diesel genset with a 12 V / 75 Ah battery.

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1. Why this document exists

Four artifacts precede this one:

Artifact	Rev	Role
Genset ECU Design Brief	J	Spec
ECU Circuit Guide	H	Per-block why/how/prove
ECU Circuit Schematics	H	15 topology sheets
PCB Design Roadmap	A	Brief → fab sequence

They were written by reverse-engineering the Kirloskar wiring diagram from memory of it, not from the document itself. Now that `GP3.8703.C4.pdf` is in hand, roughly a third of their content is wrong — not stylistically, but in ways that change the BOM, the driver topology and the pin count.

This document is the single source of truth from here. Where it contradicts an artifact, this document wins. The artifacts remain useful for the reasoning behind blocks that survived unchanged, and the Circuit Guide's "prove it" bench procedures remain valid for those blocks.

Raw extracted evidence: `refs/ecu-pinout-extracted.md`.

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2. What changed, and why it matters

2.1 Injector drive is bank-shared

The diagram traces injector high-side pin `03` to **both** cylinder 1 and cylinder 3, joined at splice V5. Pin `05` serves cylinder 2 alone. Low-side selects are individual: `73` (cyl 1), `07` (cyl 3), `29` (cyl 2).

Five pins, two banks. The artifacts specify an independent high/low pair per cylinder — six pins, three banks — which would put three boost stages on a board that needs two.

This is the conventional arrangement: the boost converter and high-side switch are the expensive, thermally-significant parts, so they are shared across cylinders that never fire simultaneously. Cylinder selection happens on the low side, which is cheap.

Consequence: two high-side bank switches, one shared boost rail, three low-side current-controlled selects.

2.2 Fuel metering unit is low-side PWM, not an H-bridge

Metering unit pin 2 sits on fused battery (20 A, `8F1`). Pin 1 goes to ECU pin `88` on a black wire — the low-side class per the page-2 legend. The ECU sinks current; it does not source it, and it never reverses polarity.

The Design Brief specifies a TLE9201SG H-bridge with one leg tied off. That works electrically but buys a reversing bridge for a solenoid that only pulls one way.

Consequence: one high-current low-side driver with current sense, closed-loop in firmware. One fewer part family to source.

2.3 Fuel level and water-in-fuel are not ECU signals

Both route through the XC11/XC1 36-pin harness to the panel — fuel level to XC pin 22 (wire `1C2`), water-in-fuel to XC pin 16/1 (wire `4B`). Neither reaches an ECU pin anywhere in the diagram. They belong to the GCU.

Consequence: two analog front-ends and two connector positions removed.

REVERSED for water-in-fuel, 8 Nov 2026 (decisions log §18). Bosch's terminal diagram wires the water-in-fuel sensor (F 002 H23 583) to ECU pin 21, `I_S_WFLS`, active high from the main-relay line. It is an ECU input again: an NPN-buffered switch input, and an amber DTC (SPN 97). Fuel level stays with the GCU.

2.4 Discrete inputs are active-high, not active-low

`AUDIO ABORT` (20), `OVERSIDE SWITCH` (24) and `IGNITION` (71) are all **red** wires — switched battery. The switch connects the pin *to battery*, not to ground.

The artifacts specify pull-up to 5 V with the switch closing to ground. That is the opposite sense, and the front-end is wrong for it: these pins see battery voltage, not 5 V, so they need a divider rated for the full 6–40 V input range plus a clamp.

The fail-safe reasoning also inverts. Circuit Guide Group D argues that normally-closed wiring makes an open harness read as a fault. With switched-battery inputs an open wire reads "not asserted" — indistinguishable from a switch that simply isn't pressed. That

argument does not hold here and should not be repeated in the new design. Detecting an open wire on these inputs requires either a pull-down and a diagnostic current source, or accepting that it is not detectable. **We accept it is not detectable** and note it in the FMEA, because changing the sense would break harness compatibility.

Pins `01` (`G_G_B AT1`) and `02` (`G_G_B AT2`) are ECU-provided return/ground legs for the switch circuits. The coolant switch (`43`) and droop switch (`23`) are cyan — digital-input class — and are conventional switch-to-return inputs.

2.5 Both CAN channels are real

Pins `55` / `77` and `87` / `86` are both drawn as twisted pairs with bidirectional markers. The artifacts treat CAN1 as a spare, DNP on the harness. In the OEM system both are wired.

Consequence: populate both transceivers and both connector positions.

2.6 Scope reductions confirmed

None of the following exist anywhere in this system: magnetic pickup unit, DEF/urea sensor, LIN bus, SCR, separate overspeed switch, barometric sensor in the harness. There is **one** catalyst temperature sensor connector (2-pin), not three EGT channels.

`OVERSIDE SWITCH` (pin 24) and `Override SS SW -S7` are one component. The Design Brief lists them as two.

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3. Safety position

> **Revised 18 Sep 2026 — second revision, and it moves in the pessimistic direction.** The genset controller has been photographed and identified. It is a **Deep Sea Electronics DSE4522 MKII AMF (India SP)**, part `4522-001-01`, serial 11021794 — **not the KG640C** that the OEM wiring diagram shows and that research memo 02 analysed. This section was built on memo 02. Its central claim is now known to be false for the device actually fitted, and the relaxation that claim bought is withdrawn.

What changed

The previous revision rested on this: *the controller senses speed from alternator frequency, not from CAN, so its overspeed trip cannot be defeated by a hung ECU.*

That is true of the KG640C. **It is not true of the DSE4522.** From DSE's own operator manual for the DSE4510/4520 MKII (document `057-260`, the 4522's siblings):

> "If the unit has been configured for CAN, compatible ECU's receive the start command via CAN and **transmit the engine speed to the DSE controller**."

and, in the engine-at-rest detection logic:

> "Engine speed is zero **as detected by the CAN ECU**"

Engine speed reaches this controller from our ECU, over CAN. The configured overspeed shutdown at 1710 rpm therefore reads a number we send. An ECU that hangs while transmitting a stale, plausible speed defeats it completely — which is precisely the failure mode this section exists to guard against.

What is confirmed, and what replaces it

No magnetic pickup — now confirmed rather than inferred. The DSE4522's full terminal list runs 1–35 and contains no speed-pickup input of any kind: 1–2 DC supply, 3–4 outputs A (FUEL) and B (START), 5 charge fail/excite, 6–9 outputs C–F, 10 sensor common, 11–13 analogue sensors, 14–17 digital inputs, 18–20 CAN, 21–24 generator voltage sensing, 25–28 mains, 29–35 CTs, charge alternator and comms. Read from the manufacturer's own terminal table, not from a diagram's silence.

An independent path does still exist — but it is a different mechanism, and a more conditional one. The module measures **generator frequency directly** at terminals 21–24, straight off the alternator's three phases. It does not use CAN to do this. The configuration sets:

Protection	Trip	Source	Equivalent speed (4 poles)
Generator **over-frequency** shutdown	**56.0 Hz** (112 %)	Alternator terminals 21–24 — **independent of CAN**	**1680 rpm**
Engine **over-speed** shutdown	1710 rpm	**CAN, from our ECU**	1710 rpm

On a 4-pole alternator at no slip, speed and output frequency are rigidly linked. The over-frequency trip fires at **1680 rpm, before** the CAN-dependent overspeed trip at 1710 rpm, and it cannot be defeated by anything our ECU does or fails to do.

Its conditions, stated plainly. It is a *generator* protection being relied on as an *engine* protection, and it inherits that mechanism's dependencies: the alternator must be excited and producing measurable output, and the AVR must be working. An excitation failure removes the protection. That is a narrower guarantee than a magnetic pickup watching the flywheel, and it should not be described as equivalent to one.

Where the chain terminates — U12 CLOSED 18 Sep 2026, and it closed badly

> **The measurement came back: the fuel relay cuts a signal. It does not remove power.**
>
> Reported from the machine alongside two other findings — the intake throttle and the pre-heat/post-heat heaters are all fed **from the battery**, not from the ECU.
>
> This is the outcome the inspection brief told the reader to flag immediately rather than save for a writeup, and the reasoning below was written before the answer was known. It stands as written; what follows is what it now means.
>
> **Both independent lines of defence are gone, for two separate reasons.**
>
> The controller has no independent speed path — it takes engine speed from this ECU over CAN (U13, closed 18 Sep). And its every protective action terminates in a relay that only signals. So a hung ECU keeps the fuel metering unit energised, and the device that is supposed to catch that is asking the hung ECU how fast the engine is turning.
>
> What survives: generator over-frequency at 56 Hz, measured from the alternator, which does not depend on this ECU for its *measurement* — but which acts through the same signalling relay. It can detect the runaway. It cannot stop it.
>
> **Confidence, stated honestly.** Brief task 1.4 exists because a normally-closed relay produces the same voltage reading with the opposite meaning, and it is not confirmed that 1.4 was performed before this result. The wording reported ("cuts signal") is a statement about function rather than a raw voltage, which suggests tracing rather than a single measurement — but that is an inference about how the work was done, not evidence. Adopting the unsafe reading costs nothing if it is wrong; adopting the safe one and being wrong costs the machine. Confirmation is still wanted, and it does not gate anything, because nothing below changes if it comes back the other way.

Every protective action this controller can take ends at the same place: it de-energises **DC Output A, terminal 3, the FUEL relay** (rated 10 A for 10 s, 5 A continuous — a relay coil driver, consistent with it operating the 70 A `-13RB1` contactor in the OEM diagram).

So the independent over-frequency path is only as good as what that relay's contacts carry. If they remove supply from the ECU and the fuel metering unit, the path works. If they merely signal a hung ECU, **the path terminates in the failure it was meant to catch**, and there is no independent protection at all.

That is U12, unchanged in substance and now the single load-bearing measurement in this section.

The position for this design

1. The ECU implements overspeed shutdown in firmware as a **primary** function, with the crank sensor as its input, at the highest scheduling priority. First line, and it is ours.
2. The controller's **generator over-frequency shutdown at 56 Hz**, measured directly from the alternator, is the **second line**. It is independent of this ECU but conditional on alternator excitation. We must not weaken it: the ECU must not latch the fuel metering unit on in a way that survives its supply being removed, and must fail safe when the ignition input at pin 71 drops.
3. A standalone MPU-driven overspeed trip module is **mandatory again**, not recommended. The relaxation in the previous revision was bought entirely by the claim that the controller had an independent *speed* path; that claim is false for the fitted device. What remains is a generator-frequency protection with an excitation dependency, terminating in a relay whose function is unverified. That is not enough to carry a 25 kVA set on its own. Two candidates are costed in memo 02 (GAC SSW675, Murphy HD9063) — that costing survives even though the memo's device analysis does not.
4. ~~**The gate: U12 must be closed by measurement before any build takes injection authority.**~~ **Closed 18 Sep 2026, in the failing direction.** The gate was written to admit the possibility that U12 came back clean. It did not. The relay signals, so line 2 detects a runaway without being able to act on it, and item 3's "mandatory" is no longer a precaution against an unverified relay — it is the only shutdown path on this machine that does not run through the ECU being protected against.
5. ~~**The trip module must break the fuel metering unit's supply directly.**~~ **SUPERSEDED 14 Oct 2026: see item 6.** The valve turned out normally open, and then breaking its supply *opens* it. The original text is kept below because the reasoning about the GCU relay chain still holds; only the contact's position changes.

This follows from U12 and is now a requirement, not a preference. A trip module that signals the same GCU relay chain inherits exactly the defect U12 just exposed. Its contacts have to sit in series with the metering unit's own battery feed — the one through fuse `8F1` — so that tripping removes fuel authority regardless of what the ECU, the controller, or the GCU relay board is doing. Both costed candidates in memo 02 are contact-output devices and can be wired this way; the costing survives, the wiring assumption behind it does not.

6. **CORRECTED 14 Oct 2026: the trip module must break the injector supply, ECU pin 04.** The old ECU's own recording (`refs/field-evidence-2026-10-14.md` §3a; decisions log §14) shows a **normally-open** metering valve: its current falls as rail pressure rises, and rises to 1684 mA to stop the engine. With such a valve, removing its supply gives *full* flow. Item 5's arrangement would leave the pump delivering to the rail, with the rail limited only by its relief valve, and **would not stop a runaway at all.**

On this board, **all injector energy enters on pin 04** (`V_BAT_1R` → `VBAT_INJ`): - it feeds the boost converter that charges the 100 V reservoir - it feeds the two battery high sides that carry the hold current

Nothing else powers an injector. So the trip module's contacts go **in series with the pin 04 feed, after its 10 A loom fuse.** When it trips: - **Hold current stops at once,** because the battery path is gone. - **The boost reservoir runs down within a few injections.** It is 41 µF and loses about 8.3 V per peak phase (`gen_sheet_injector.py`, from `boost_converter.cir`), so roughly a dozen peak phases empty it from 100 V. That is about 0.3 s at 1500 rpm, and it is an estimate from those two numbers, not a simulation. - **Pin 21 (logic), pin 06 (EGR) and the metering valve's own feed are untouched.** The ECU stays alive to log the trip and report it over CAN.

The contact carries real current. The pin 04 feed is 10.98 A peak and 3.8 A RMS, behind a 10 A loom fuse (`gen_sheet_power_input.py`). If the module's own contact isn't rated for that, it drives a **run relay** instead: - a 12 V automotive relay with a normally-open contact rated ≥ 30 A, and coil suppression - its coil is fed through the module's "healthy" contact, so the relay is **energised to run** - then a trip, a module power loss, or a broken coil wire all stop the engine

What the ECU sees: pin 04's voltage is already measured on the harness side (`V_BAT_1R_SNS`, ADC). Pin 04 dropping while pin 21 stays up *is* a trip, so the firmware can recognise and report one without a new input. Whether that makes the supervised `TRIP_LOOP` input unnecessary is **open**, for the owner. The loop also reports a cut wire to the module, which the voltage alone doesn't.

7. **SUPERSEDES ITEM 6, 8 Nov 2026 (Bosch pinout, decisions log §18): the trip module opens the main relay's coil feed.** Under Bosch's wiring pins 04/06 come from the main relay's contact and are the board's *only* supply, so breaking pin 04 now drops the ECU as well. Item 6's "the ECU stays alive to log the trip" no longer holds. The trip loop had no Bosch pin and is gone.

The module's contact goes **in series with the relay coil's Term. 30 feed**, between the coil's fuse and relay terminal 86. When it trips, the relay opens and takes everything with it: - **The injectors**, on 04, both their hold path and the boost converter's input. - **The EGR**, on 06. - **The ECU**, also on 04. - **The metering unit and the buzzer relay**, on the main-relay line.

No injection is possible with 04 dead. The normally-open metering valve going to full flow therefore feeds a rail with no outlet but its relief valve, which is item 6's reasoning applied one step earlier.

- **The contact carries only the coil current**, about 84 mA, so the module's own contact is enough and item 6's run relay is not needed.
- **The ECU cannot defeat it.** The ECU switches only the coil's low side (pin 50, and the wake FET from pin 71). With the coil's feed open, neither can close the relay.
- **What the ECU sees:** nothing. It loses power. The GCU sees the J1939 traffic stop and alarms on it. A logged trip is what this gives up. The module's own indication, and the GCU's, record it instead.

The owners may of course revisit item 3 with evidence. What they should not do is inherit the previous revision's conclusion, because the fact it was built on turned out to describe a controller that is not on this machine.

What this ECU must do about the trip module

Item 6 puts the trip module's contacts in series with the injector feed, pin 04 (item 5, the metering unit's feed, is superseded). That interface is the module's, not ours, but it constrains this design in three ways.

We must not be able to defeat it. The trip module removes pin 04. Nothing on this board may power the injector stage from anywhere else: not pin 21's `VBAT_PROT`, not pin 06's `VBAT_EGR`, and not through a diode-OR or a protection part's leakage path. Today the schematic has none. `VBAT_INJ` is fed only from `V_BAT_1R` through Q103. The boost controller's own supply (`VDD_BAT` from `VBAT_PROT` through R405) and the high-side bootstraps (from `12V_GATE`) come from pin 21. They carry milliamps of control current, not injector current. With pin 04 open they keep the drivers alive but give them nothing to switch. That property must survive every revision. **Any path around the trip contact silently restores the failure the module exists to remove.** Any such path would be a wiring error that silently restores the failure mode the module exists to remove.

We should sense its state, and treat it as an input rather than a surprise. A trip that removes fuel while the ECU is still commanding injection produces a fault the ECU should recognise and report over CAN, rather than reading it as a fuelling anomaly and compensating.

The cheapest form is one discrete input sensing whether the module's contacts are closed. That is a new connector position and a new front-end, and it does not exist in the OEM harness — a deliberate addition, listed under §7.

We must fail safe into it, not against it. The module acts on overspeed sensed by its own magnetic pickup. The ECU's own overspeed shutdown (item 1) acts on the crank sensor. These are independent measurements of the same quantity, and they will disagree near the threshold. The ECU must not treat its own reading as authoritative to the point of trying to keep running through a trip.

Not yet decided: whether the module's pickup shares the flywheel target with our crank sensor or gets its own. Sharing is cheaper and physically simpler; separate targets preserve the independence that is the entire point of fitting the module. This is a mounting question for whoever fits it, and it should be answered before the module is ordered rather than after.

What this section still does not address

A diesel that begins burning its own lubricating oil — through a failed turbocharger seal or crankcase fumes drawn into the intake — has a fuel supply that **none of the three lines above can interrupt**. Every one of them cuts diesel or electrical power. The standard countermeasure is a mechanical or pneumatic **air-intake shutoff valve**, and no such device appears anywhere in this design, this register, or §9's out-of-scope list. It was never considered rather than considered and rejected.

This is recorded here, unresolved, because it is a decision for the machine's owners: the part is mechanical, not electronic, and fitting it is not within this PCB's scope. It should not be allowed to stay invisible.

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4. ECU pin map

Since 8 Nov 2026 the pin map is Bosch's terminal diagram for this ECU (`refs/bosch-pinout-md1cs162.md`, 46 pins, decisions log §18). It replaced the KOEL wiring diagram below where they disagree, and they disagree on power. The tables in this section are kept as written, with the Bosch changes in this table:

Pin	Bosch	What the board does
21	`I_S_WFLS`, water in fuel	Switch input, active high. **Not** a battery feed: there is none
04 / 06	`V_V_BAT_1R/2R`, main relay contact	The board's only supply. 04 feeds the electronics chain (5 A fuse, EMI filter, TVS, reverse FET, buck) and the injectors; 06 the EGR
50	`O_S_MRLY`	Closes the board's own supply relay; held through the after-run

Pin	Bosch	What the board does
71	`I_S_T15`	Ignition, and wakes the board: a second FET on pin 50
01 / 02	`G_G_BAT1/2`	Power ground
09	`I_A_SYNC`	GCU synchronisation input, 0–10 V divider to ADC
14	`I_S_DIA`	Diagnosis request, contact to ground
25	`I_S_STST`	Stop / emergency switch, contact to ground: SHUTDOWN

Confirmed pins from the wiring diagram. **44 of 94.** See §8 for the remaining 50.

Power

Pin	Signal	Front-end
21	~~Battery + feed~~ **Water in fuel (Bosch)**	~~Main input: EMI filter → fuse → TVS → reverse-battery FET~~ That chain now takes pin 04
04	`V_V_BAT_1R` (10 A fused)	**120 k / 10 k** divider → ADC; **injector feed** `VBAT_INJ` since 8 Oct 2026 (POWER-PATH, decisions log §2a, §5); **the electronics feed** since 8 Nov 2026
06	`V_V_BAT_2R` (10 A fused)	**120 k / 10 k** divider → ADC; **EGR feed** `VBAT_EGR` since 8 Oct 2026 (POWER-PATH, decisions log §2a, §5)
09	~~`SYNCHRONIZATION GROUND`~~ `I_A_SYNC` (Bosch)	Analog input from the GCU

The battery-sense divider is **120 k / 10 k, not the 75 k / 10 k** inherited from the earlier artifacts. That value was sized for a 5 V ADC and delivers 4.69 V at a 40 V input — above the S32K148's 3.3 V analog supply. The corrected divider peaks at 3.07 V and still resolves 573 ADC counts at the 6 V cranking dip. Verified in `sim/blocks/battery_sense.cir`.

Ratiometric analog — 10 k / 16 k divider, 22 nF shunt, 3.3 V clamp, rail measured and corrected in firmware

Revised 14 Sep 2026. This section previously specified *1 kΩ series, 100 nF shunt, ADC ref tracks* `5V_SENSOR`, which contradicted the simulated front-end and cannot be built on this MCU. Both halves were wrong, and for the same reason the 75 k / 10 k battery divider was wrong — they were written for a 5 V part:

- `VREFH` **cannot track** `5V_SENSOR`**.** The S32K148's ADC reference may not exceed its 3.3 V analog supply. A reference at 5 V is not an option on any 3.3 V device.
- **1 kΩ in series scales nothing.** With no division, a sensor at 4.5 V arrives at a 3.3 V pin. The series resistor is a fault-current limiter, not an attenuator.

So the signal must be divided, and the question becomes how to keep the *ratiometric* property after dividing it.

Decision: divide, then measure the rail and correct in firmware. `5V_SENSOR` gets its own ADC channel. Every ratiometric reading is computed as a fraction of the measured rail rather than of an assumed 5.000 V, which is what the sensor's output actually represents. Cost: one ADC channel, zero analog parts. The channel budget has room — see the sensor reference.

Rejected alternative: dividing `5V_SENSOR` into `VREFH` by the same 16/26 ratio. It fits numerically (5 V × 16/26 = 3.08 V, inside the 3.3 V limit) and gives true ratiometric conversion with no firmware arithmetic. It is rejected because `VREFH` is shared by every channel on that ADC, so it would silently make **battery sense and the NTC channels ratiometric to the sensor rail as well** — both of which need an absolute reference. It would also need a buffer, since a resistive divider is too noisy and too high-impedance to drive a reference input.

Single-ended or differential is decided by the ground, not by the channel. Where the sensor's return is shared, the other sensor's current develops an offset across the harness resistance that a single-ended input reads as signal: `sim/blocks/sensor_differential.cir` measures 12 mV of bias at 1 Ω of return resistance against 0.13 mV differential. Research memo 09 found both Deep Sea Electronics and SEDEMAC treat this as a first-class problem. So:

- **Shared ground (pins 34, 36) → differential front-end.**
- **Dedicated ground (pin 08) → single-ended is sufficient**, because there is no other sensor's current in that return. This mirrors the OEM's own ranking: they gave rail pressure a dedicated return precisely so it would not need one.

Pin	Signal	Group ground	Front-end
11 / 41	Boost: 5 V excitation / pressure signal	34	Differential
79	Boost: temperature signal — **topology unresolved, see §8**	34	Differential
34	Sensor ground — boost + coolant	—	—
33	Coolant temperature (NTC, pull-up divider)	34 (spliced)	Differential
15 / 37	EGR position: 5 V / wiper	36	Differential
36	Sensor ground — EGR + oil pressure	—	—
39 / 80	Oil pressure: 5 V / signal	36 (spliced)	Differential
32 / 35	Rail pressure: 5 V / signal	08	Single-ended
08	Sensor ground — rail pressure, dedicated	—	—

Pin	Signal	Group ground	Front-end
—	`5V_SENSOR` rail monitor	—	Divider → ADC, for the correction above

Sensor-ground grouping mirrors the OEM harness exactly: `34` boost+coolant, `36` EGR+oil, `08` rail alone, `30` crank alone, `44` cam alone. Rail pressure gets a dedicated return because it is the most accuracy-critical analog channel on the engine.

Speed

Pin	Signal	Front-end
52 / 74	Crank VR: Frequency I/P High / Low	Differential adaptive-threshold conditioner → timer capture
30	Crank sensor ground	Shield terminates here, ECU end only
45 / 46	Cam Hall: 5 V / Frequency I/P	**This row was wrong — see the correction below**
44	Cam sensor ground	—

Discrete inputs

Pin	Signal	Sense	Front-end
43	Coolant switch	Switch-to-return	Pull-up + RC debounce
23	Droop switch	Switch-to-return	Pull-up + RC debounce
01 / 02	`G_G_B AT1` / `AT2`	—	Switch return legs — but see note below, these may be power grounds
20	Audio abort	**Active-high**	47 k / 68 k bias + 3.0 V zener clamp + 220 nF
24	Override SS	**Active-high**	same
71	Ignition	**Active-high**	same

The active-high front-end is not a divider, and cannot be. Reading logic high at a 6 V cranking dip needs a divider ratio above 2.31/6 = 0.385; staying under 3.3 V at 40 V needs it below 3.3/40 = 0.0825. Those constraints do not overlap, so no fixed ratio works at both ends. The zener sets the clamp level and the divider only biases into it, which makes the logic level flat to within 92 mV across the whole range instead of tracking the battery. Verified in `sim/blocks/discrete_input.cir`.

Pins 01/02 need checking. Memo 03 infers from the connector drawing that pins 1, 2, 5 and 6 are large power contacts. Pins 05 and 06 fit that — injector high side and a battery rail. Switch return legs would not. The likelier reading is that `G_G_B` denotes battery ground and these are the ECU's main power grounds, which would also explain why no power-ground pin appears anywhere in the extraction. Queued as a continuity measurement.

Outputs

Pin	Signal	Driver
03	Injector high side — bank A (cyl 1 + 3)	Boosted high-side switch
05	Injector high side — bank B (cyl 2)	Boosted high-side switch
73 / 07 / 29	Injector low side — cyl 1 / 3 / 2	Current-controlled low-side, peak-and-hold
88	Fuel metering unit PWM	High-current low-side + current sense
59 / 81	EGR High / EGR Low	H-bridge + position feedback on pin 37
50	Main relay	Low-side FET + flyback
69	Buzzer relay	Low-side FET + flyback

`Cam front-end (pin 46) — this row was wrong, CORRECTED 19 Sep 2026`

The outputs table said "pull-up to `5V_SENSOR` → timer capture" for the cam frequency input. A Hall sensor pulled up to the 5 V sensor rail swings to 5 V, and the S32K148 is a 3.3 V part. Taken literally, that drives an MCU pin above its absolute maximum.

It survived because **nothing simulated it.** There is a `vr_conditioner` block for the crank sensor and no cam equivalent, so the mechanism that catches this class of error never looked. Found by the pin-map work, on the way to something else.

This is the second front-end here specified in a way that would have destroyed a pin. The first was a divider on the battery sense — caught by `battery_sense.cir`, precisely because a block existed.

Now built: `sim/blocks/cam_frontend.cir` divides the 5 V swing into the 3.3 V domain and verifies it across the rail's own range — 3.072 V nominal, 3.147 V at 5.25 V high line, 2.769 V at the 4.5 V fault floor `sensor_rail.cir` established, where it still reads a valid logic high. All clear the 3.6 V absolute maximum. It also checks a 40 V harness short, which clamps the pad to 3.308 V with fault current held to 3.7 mA.

`Nothing on this board protects the buck against a NEGATIVE input — GAP, found 19 Sep 2026`

`sim/blocks/negative_pulses.cir` simulated ISO 7637-2's negative transients for the first time, which became possible only once the TVS stopped being modelled as a single diode.

The LM5164's VIN pin is rated −0.3 V to 100 V. The negative limit is not a mirror of the positive one, and nothing in this design was ever sized against it. Read from TI's own absolute-maximum table.

With the reverse-battery FET assumed still conducting — the honest assumption, since no turn-off delay is sourced anywhere and pulse 3a is only 0.1 ms wide:

Pulse	Buck VIN	Past the −0.3 V rating by
1 (−150 V / 10 Ω / 2 ms)	**−40.7 V**	135×
3a (−220 V / 50 Ω / 0.1 ms)	**−27.8 V**	93×

The bidirectional TVS clamps correctly. It clamps to its own breakdown, and −38 V is a perfectly good clamp that is still two orders of magnitude outside what the buck tolerates. **This is a missing component, not a mis-sized one** — the chain has no series element or shunt diode addressing a negative excursion.

And the reverse-battery stage cannot be assumed to cover it. If its controller opens fast enough the buck is isolated almost perfectly; if not, the FET conducts and the buck sees the full clamped negative voltage. `reverse_battery.cir` models instant turn-off and its own RESULT NOTE warns it "will always look perfect in reverse". **No controller part is chosen and no turn-off delay is sourced**, so both bookends are simulated and neither is claimed. That delay is now a load-bearing number, and this decision should be taken with the controller selection rather than separately.

`Every driver gate is held off by hardware, not by firmware — REQUIRED 18 Sep 2026`

Research memo 11 establishes the hazard with the MCU's own datasheet numbers rather than by argument. Two facts combine badly:

- **S32K148 GPIOs are high-impedance out of reset**, until firmware configures them. High-impedance on a gate driver is not "off" — it is undefined, and the gate can be brought up by leakage or by capacitive coupling from a switching node.
- **There is a band where nothing internal resets the part.** Low Voltage Reset always forces a reset at 2.50–2.7 V, but Low Voltage Detect at 2.8–3.0 V only does so if firmware has set `LVDRE`. Below the datasheet's own 2.97 V correctness guarantee and above LVR, the core may execute wrong instructions **with its outputs still driven**.

The pin that makes this concrete is **88**. The fuel metering unit is fed from battery through fuse `8F1` and switched low-side by this ECU (§2.2). A gate floating high during reset energises the metering unit with no firmware in control of it — and, since U12, with nothing downstream able to remove its power either.

Therefore, as a binding requirement on every driver gate in the table above:

1. A **gate–source pulldown** at each gate node, sized by the FET actually chosen rather than fixed here:

> `Rpd < Vgs(th) / (Crss · dV/dt)`

The dV/dt in that formula is the weakest number in this requirement, and this document previously overstated its standing. It read "2.8 V/µs from `injector_turnoff.cir`", which is a citation that file explicitly refuses: its own comments say reporting a dV/dt from that model "would be exactly the kind of promotion-to-sourced the project's own discipline rules out." The figure is memo 10's **inferred** estimate — 100 V in ~36 µs — and `supervisor.cir` imports it as an assumed input waveform, labelled INFERRED there too. Every file closer to the number than this one hedges it; only the spec stated it flatly, beside genuinely simulated figures, where it read as one of them.

It is also an average, not a peak, and the error runs in the unsafe direction. 100 V over 36 µs is the mean slope of the whole turn-off. Miller coupling happens during the plateau, where a real FET slews far faster. The requirement's sensitivity to that, at Crss = 500 pF:

dV/dt	gate through 470 Ω	Rpd actually needed
2.8 V/µs (the assumed average)	0.66 V — ok	< 714 Ω
10 V/µs	2.35 V — over threshold	< 200 Ω
28 V/µs	6.58 V — over threshold	< 71 Ω

At 200 Ω the gate driver would need an output impedance near 10 Ω; at 71 Ω a passive pulldown cannot be driven at all. **So the 470 Ω answer survives only at the assumed average**, and a modest factor on the real slew rate removes the passive-pulldown topology entirely rather than merely resizing it.

A second assumption pulls the other way. The 470 Ω also assumes Vgs(th) = 1.0 V, the conservative end of a 1.0–2.5 V class range with no part chosen. Read at 2.5 V the same sweep gives **1000 Ω and a ≤52.6 Ω driver**. So:

Assumption	Direction	Effect on 470 Ω
Vgs(th) 1.0 V vs 2.5 V	permissive if wrong	470 Ω is **pessimistic** — costs drive current, not safety
dV/dt 2.8 V/µs (an average, not a plateau slew)	**unsafe if wrong**	470 Ω is **optimistic** — a real slew of 10 V/µs demands 200 Ω

The slew-rate assumption is the larger lever and the dangerous direction. **Treat 470 Ω as a demonstration, not a value to build to:** what it actually shows is that a passive pulldown alone lives in a narrow window bounded by two numbers nobody has measured. That conclusion holds whichever way either assumption lands, and it is the argument for the active clamp — which `sim/blocks/supervisor.cir` has already validated, and which research memo 12 recommends building discretely regardless of the driver-architecture decision.

At a 1.0 V threshold that is **under 1.79 kΩ for Crss = 200 pF, 714 Ω at 500 pF, 357 Ω at 1000 pF**. `sim/blocks/supervisor.cir` claim 6 sweeps this directly and lands on **470 Ω** at Crss = 500 pF — twenty times smaller than the figure this requirement first carried.

That value constrains the gate driver, and the two cannot be specified apart. At 10 kΩ the pulldown was a perturbation on the driver. At 470 Ω it is the bottom half of a divider with the driver's own output impedance:

> `Vgs = Vdrv · Rpd/(Rpd + Rdrv)` — holding 9.5 V across 470 Ω needs **Rdrv ≤ 24.7 Ω**

It also costs **21.3 mA per gate and 1.28 W across all six**, continuously, whenever the FETs conduct. Stating the pulldown as a single number hid both the coupling and the power.

This reopens a topology question memo 11 closed on the bad arithmetic. The memo considered an active clamp and dismissed it. A passive pulldown strong enough to reject Miller coupling at 500 pF is strong enough to fight its own driver and burn over a watt doing it — which is the trade an active clamp exists to avoid, and which an integrated driver in the MC33814 class handles internally. That decision is now coupled to the driver-architecture decision below and should be taken with it.

This requirement said "10 kΩ" when first written on 18 Sep 2026, and that was wrong. Memo 11 §4 computed the Miller-coupled current correctly at 140–560 µA and then mis-multiplied, reporting 5.6 mV across 10 kΩ where the answer is 5.6 V. `sim/blocks/supervisor.cir` was built to check that paragraph and fails it at 50 pF, the low end of the memo's own range. The memo now carries the correction inline. The error is recorded rather than quietly patched because it survived a review, a spec commit, and was caught only by simulating the claim. 2. A **windowed supervisor** monitoring `3V3_MCU`, tripping at **~3.05–3.10 V**, above LVD's 3.0 V maximum, so the trip does not depend on whether firmware ever configured `LVDRE`. Its watchdog window must close well inside one cylinder interval (26.67 ms at 1500 rpm). 3. The supervisor's reset, inverted, driving a **kill transistor at each protected gate**, so the safe state is enforced independently of GPIO configuration. A plain timeout watchdog is defeated by a stuck loop that still kicks it; the window is what closes that.

With the normally-open valve (14 Oct 2026), the kill still stops the engine. It also holds the metering driver's gate off, which leaves the valve fully *open*, but the same kill holds every injector gate off, so nothing is injected and rail pressure goes to the relief valve. The safe state is "no injection", not "no rail pressure".

This is the ECU's own fail-safe and it is **separate from the standalone trip module** of §3. The trip module removes fuel authority from outside the board; this keeps the board's own outputs defined while it still has power. Neither substitutes for the other, and since U12 neither is optional.

Open, and blocking the detail rather than the requirement: the physical `PTxx` assignment for pins 88, 73/07/29 and 03/05 is a phase-2 task (§5), so the per-pin reset pull defaults cannot yet be looked up. The requirement above does not depend on them — it exists precisely so that the answer does not matter — but the analysis cannot be finished until the pin map is fixed.

`Input protection — RESOLVED 20 Sep 2026, and it moved three other parts`

Two decisions, taken together because they are coupled.

1. TVS: SMBJ33CA → SMCJ43CA. The standoff had to clear the 40 V clamped load dump so the part stops conducting during a normal event — that is what `load_dump`'s failure was actually saying. 43 V is the smallest standard bin above 40 V, and smallest matters because clamping voltage rises with the bin and everything downstream sees it.

The package changed too, and that was forced by simulation rather than chosen. At the 600 W SMBJ size, pulse 2a clamped at **88.6 V** against the buck's 100 V absolute maximum — 11 % margin, too thin to design to. The 1500 W SMCJ part has half the dynamic resistance at the same standoff and clamps at **73.3 V**, restoring 1.37× margin.

Result: `load_dump` now draws **0.000 A and absorbs 0.000 J** — the TVS simply stays off through the dump, which was the predicted fix and is now the simulated one.

2. A negative clamp diode at the buck input. The TVS holds the node at its own −47.8 V breakdown, which is a correct clamp and still two orders of magnitude outside the buck's **−0.3 V** rating. A 60 V / 20 A-class Schottky, anode to ground and cathode to VIN, conducts long before the TVS does:

Pulse	Buck VIN before	After
1	−40.7 V	**−0.53 V**
3a	−27.8 V	**−0.39 V**

The residual is honest and still fails its check. 0.53 V is 1.76× the −0.3 V rating, not inside it. No silicon diode reaches 0.3 V at the ~15 A this fault delivers. Closing the last 0.23 V needs series impedance — the EMI filter's ferrite sits upstream and is not modelled here — or a judgement that a sub-microsecond 0.2 V excursion past a DC absolute maximum is acceptable. **That is a question for the part's manufacturer, and it is now a bounded one rather than a missing component.**

What this moved elsewhere: the reverse-battery P-FET sits downstream of the clamp, so it now needs **100 V class, not 60 V** — its third revision, each time because the clamp moved. `transient_clamp.cir` now carries an explicit check stating the downstream requirement, so the next move does not have to be noticed by a human reading three files.

`The EGR driver must decode IN1/IN2 internally — REQUIRED 19 Sep 2026`

`sim/blocks/egr_hbridge.cir` argued that no IN1/IN2 combination could short a leg, because each input line drives a *diagonal* pair. A review falsified it against the block's own wiring: IN1 commands leg-A-high and leg-B-low, IN2 commands leg-B-high and leg-A-low, so asserting **both** turns all four switches on and shorts **both legs** rail-to-ground at once. Simulated at **270 A**, matching the hand figure of two legs of 2 × 50 mΩ across 13.5 V.

The reasoning failed by checking each input line alone and never asking what the two do together.

Nothing on this board currently prevents that state. The gate pulldowns hold both lines low at reset, which is correct and not the issue — the exposure is any fault or firmware error that drives both high.

The requirement: the EGR driver must decode IN1/IN2 so that the 11 state means brake or coast and never shoot-through. Integrated H-bridge drivers do this internally, and that now counts as a reason to prefer one here — note that memo 12's driver decision covered the *injectors* and explicitly did not address EGR. If the driver is discrete, the interlock has to be built, and **a per-leg dead time is not sufficient on its own**: this fault is a logic state, not a timing overlap.

RESOLVED 20 Sep 2026. The same bridge driven through a decoded DIR/PWM interface, with **both** inputs asserted — the combination that shorts the naive arrangement — draws **4.35 A**, which is the motor's own stall current, not a short. 270 A against 4.35 A. The fault state does not exist under decode; it is not merely made less likely.

`egr_hbridge.cir` now simulates both, and the naive wiring is kept as **evidence of why the requirement exists** rather than as a live failure — the same treatment `battery_sense.cir` gives the divider it rejected.

What to draw: an integrated H-bridge with a decoded DIR/PWM (or equivalent non-shoot-through) input interface, current limiting, and headroom over the actuator's stall. `egr_hbridge.cir` puts stall at **2.6–6.4 A** across a plausible armature resistance range, with no actuator chosen — so specify a part with comfortable margin over 6.4 A and confirm against the actuator once U8-class information arrives.

Candidate parts are not named here: none has been checked against a retrieved datasheet, and this project's rule is that an unread figure does not become a design number.

`Driver architecture — DECIDED 19 Sep 2026, research memo 12`

Discrete power MOSFETs with a gate-driver IC for the injector banks, keeping the 100 V boost rail. An integrated smart low-side switch for the metering unit. An active kill-clamp at every gate of both, built discretely.

The decision was forced by one confirmed number. The two integrated injector-driver ICs that fit this design's channel count and current class both cap their boost rail **below the 100 V already chosen**: MC33816's `VBOOST` absolute maximum is **72 V** (NXP datasheet Rev 10.0), and L9781's tank voltage is up to 80 V (ST data brief DB3720, supplied by the owner 8 Oct 2026, after the datasheet failed retrieval five times).

Adopting either would force the boost rail down and re-derive three already-passing blocks. Memo 12 re-did that arithmetic: at 72 V the peak-phase ramp stretches from 38 µs to **53 µs**, which still fits a pilot injection but consumes half of it.

And neither part removes the work. Both are pre-drivers, not integrated power stages — external MOSFETs, external boost inductor and diode, and external sense shunt are required either way. MC33816 additionally brings its own microcode toolchain. MC33814 is the wrong class outright: 1.3 A continuous against an 18 A requirement, and no boost stage at all.

On the coupled clamp question, the integrated route relocates the problem rather than removing it — and does so in a form worth copying. MC33816's high-side pre-driver is *actively forced low* on reset, falling back to a weak 500 kΩ–2 MΩ resistor only once its bootstrap capacitor is exhausted. That is the same two-tier structure `supervisor.cir` already validated, on a different bias rail, and with no datasheet statement about how fast that bootstrap decays under brownout — where `supervisor.cir` claim 4 gives a timed guarantee. **So build the active clamp discretely regardless.** This was never a question the IC-versus-discrete decision could have answered differently.

What would change this: a retrieved L9781 datasheet showing a tank voltage at or above 100 V, or a decision to accept a ≤72 V rail on other grounds.

One sourcing item escalated and unresolved: MC33816AE carries a distributor last-time-buy date of 06/08/2027, which needs a human with distributor access. The L9781 item is closed: the owner supplied ST's data brief on 8 Oct 2026, confirming the 80 V tank ceiling.

CAN

Pin	Signal
55 / 77	CAN channel 1 — H / L
87 / 86	CAN channel 2 — H / L

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5. MCU selection

Recommendation: NXP S32K148. This resolves the contradiction in the artifacts — Design Brief §03 specifies MPC5744P while its own §02 block diagram and §06 sensor table still say S32K148 and FTM.

The argument for MPC5744P was angle-synchronous injection timing needing the eTPU2.

> **Corrected 2026-09-13** after research memo 05 challenged the original reasoning and was right to. This section first argued that constant 1500 rpm operation relaxes the timing problem because a crank degree is 111 µs and a 60-tooth wheel gives an edge every 667 µs. That checks the wrong variable. Crank-edge *capture* rate was never the binding constraint in either case — at 6000 rpm a 60-tooth wheel produces an edge every **83 µs**, which is about **6,700 CPU cycles** at 80 MHz against an NVIC latency of roughly 12. The comparison was never close, so constant speed cannot be what makes it comfortable. The reasoning below replaces it.
>
> **Re-corrected 18 Sep 2026.** This box said "an edge every 28 µs, which is over 2000 CPU cycles". 28 µs is the **one-degree-of-crank** period at 6000 rpm (10,000 µs ÷ 360), not a tooth-edge period — a unit conflation, off by a factor of six, sitting inside a box that was itself written to fix an earlier reasoning error. A 60-tooth wheel at 6000 rpm gives 166.7 µs per tooth and 83.3 µs per edge counting both polarities.
>
> The conclusion is unaffected and in fact strengthened: the correct figure gives *more* margin, not less. It is corrected because a wrong number that happens to support the right answer is still a wrong number, and this one would have been inherited by anyone reasoning forward from it.

What the eTPU2 actually buys. It is a co-processor that autonomously schedules and re-schedules many angle-domain output-compare events across a wide and *rapidly changing* speed range, without consuming the main CPU's interrupt budget and without the core's scheduling jitter — cache effects, higher-priority ISRs, RTOS latency — reaching the injection timing chain. That matters when rpm sweeps 800–6000 with tight transient response and pilot/main/post multi-pulse strategies all landing within microseconds of a moving target angle.

Why we don't need it. This set governs to a fixed 1500 rpm, and the pin map in §4 shows a single main injection pulse per cylinder per cycle — one bank high-side plus one low-side select, with no multi-pulse rate shaping in scope. The constraint that does bind is **injector pulse-width resolution**, which sets fuel-quantity metering accuracy and is independent of crank speed. FlexTimer output-compare runs off the bus clock: one tick is 12.5 ns at 80 MHz, 8.9 ns at 112 MHz HSRUN. Against a single-pulse metering requirement on the order of 1–4 µs, that is roughly 100–450× headroom.

The S32K148 also carries 2× Programmable Delay Blocks, which trigger ADC conversions or timer events at a defined delay from a capture event in hardware. That recovers part of what the eTPU2 offers — jitter isolation on the timing chain — without the eTPU2.

Caveat carried forward: the 1–4 µs resolution figure is a general diesel-injector engineering figure, not this injector's datasheet. Revisit if U5 closes with a part demanding materially finer control, or if multi-pulse injection ever enters scope.

What the S32K148 buys in exchange:

- Free S32 Design Studio; GCC ARM toolchain; cheap CMSIS-DAP/J-Link probes instead of a Power Architecture Nexus pod
- Stock KiCad symbol availability, or trivially built from NXP's own EVB files
- Real availability in small quantities in India
- 3× FlexCAN (we need 2); dual 12-bit ADC against a demand of ~14 channels — 8 sensor inputs, 2 battery-rail senses, and 4 current-sense returns from the injector, metering and EGR drivers; FlexTimer instances well beyond the ~10 timing channels required (crank capture, cam capture, 3 injector low-side, 2 injector high-side, metering PWM, 2 EGR bridge PWM)
- No eTPU2 microcode, which is a specialist skill and a hiring risk on a small team

Confirmed in Phase 1 — see `docs/research/05-mcu-selection.md`. Orderable part **FS32K148HAT0MLQT**, 144-pin LQFP, 80 MHz RUN / 112 MHz HSRUN grade, −40…+125 °C. In stock at DigiKey India at roughly ₹1,900–2,000/unit. The S32K146 fallback was checked and is in worse supply shape (backordered, 12–52 week spread), so it is not needed.

Two items deferred to Phase 2: confirming how many ADC channels the 144-pin package actually breaks out (needs the reference manual's pin-mux table, not the datasheet), and pad-by-pad verification of the SOT486-2 package drawing before the footprint is locked.

Pin map done, 19 Sep 2026 — see `docs/pinmap.md`. The first of the two deferred items above is closed: the Reference Manual's own embedded `S32K148_IO_Signal_Description_Input_Multiplexing.xlsx` (not the datasheet) gives 128 GPIO-capable pins on the 144-pin package, of which this design commits 37, including all 94-way connector signals with a confirmed identity (spec §4). It also settles §4's own deferred fail-safe detail: all six gate pins (88, 73/07/29, 03/05) are confirmed high-impedance with no pull at reset, exactly the condition §4's hardware kill-clamp was already designed against. One new finding surfaced in the process: pin 46's cam front-end, as specified above, would put a 5 V swing on a 3.3 V MCU pin — flagged there, not fixed here. Package pad-pitch verification remains open.

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6. Power architecture

Unchanged from the artifacts in topology; the reasoning there is sound.

Stage	Function
EMI input filter	Common-mode choke, CISPR 25 conducted margin
Reverse-battery	Ideal-diode controller + P-FET, < 0.3 V drop
Transient clamp	Bidirectional TVS, ISO 7637-2 pulses incl. load dump 5b
Buck pre-regulator	6–40 V → 5 V, wide-input synchronous
`5V_SENSOR`	PTC-protected per sensor group, isolated from MCU rails
`3V3_MCU`	LDO off 5 V
Supervisor	Windowed watchdog + brown-out reset

Input range 6–40 V: 6 V covers the cranking dip on a 12 V system, 40 V bounds the clamped load-dump transient.

Bosch wiring, 8 Nov 2026: there is no permanent battery feed. The chain above starts at pin 04, the main relay's contact. Pin 71 (ignition) closes the relay through a wake FET on pin 50, and the MCU holds it through the after-run: it stops the engine, writes the fault memory, then releases it. Pin 50's turn-off clamp is a TVS to ground. A diode to the board's supply would feed the board through the always-live coil (`sim/blocks/main_relay.cir`).

Injector boost rail is new relative to the power chain as drawn: a separate boost stage generating the injector opening voltage, with its own reservoir bank sized for the pulsed draw of two banks. Target voltage is set by the injector part — see §8.

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7. Deliberate deviations from the OEM design

Recorded so they are choices, not drift:

1. ~~**Connector split.** The OEM uses one 94-pin connector. We use functionally-split sealed connectors. This simplifies harness build and field diagnosis, at the cost of needing an adapter harness to mate with an unmodified Kirloskar loom. That adapter is a deliverable, not an afterthought.~~

REVERSED 22 September 2026, by the owner. The ECU carries the **OEM connector described in the field evidence** — Bosch `1 928 405 192` / `1 928 405 194`, **code C**, 94-way, EDC17 family (`refs/field-evidence-2026-09-18.md` §2b) — and plugs straight into the unmodified Kirloskar loom. **No adapter harness.** This is no longer a deviation from the OEM design.

Consequences, all already reflected elsewhere: the board-side land pattern is now load-bearing (field brief task 5.3, and a Bosch distributor quoting both numbers, since neither returns a catalogue hit); **code C** keying must match exactly; and the harness-side questions — which cavity is the power ground (task 5.1), and a pin and wire for the new `TRIP_LOOP` (task 5.2) — are answered in the OEM loom itself rather than designed into an adapter. 2. **Onboard barometric sensor.** Not present in the OEM harness. Added on-PCB behind a vented port; costs nothing in harness terms and improves fuelling correction. 3. **Independent overspeed path.** Added, per §3. The OEM does not have one.

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8. Unknowns register

Nothing in this list may be guessed into copper. Each needs an answer before the sheet that depends on it is drawn.

Status updated at phase 1 exit, 14 Sep 2026. Three states, and the middle one carries weight: **Open** means no answer yet; **Answered** means phase-1 research produced an answer that is good enough to design against but is *inferred*, not confirmed, and still carries the upgrade path in its memo; **Closed** means confirmed outright. Nothing here is Closed yet — every confirmation on this list needs the machine.

#	Unknown	Status	Blocks	How to close
U1	ECU connector identity	**CLOSED 18 Sep 2026 — Bosch**, part numbers `1 928 405 192` and `1 928 405 194`, `code C` mechanical keying. 94 cavities: power pins 1–8 in their own chamber, then rows 9–28, 29–50, 51–72, 73–94	—	Closed by markings on the part. Note: an adapter housing **must be code C** or it will not mate

#	Unknown	Status	Blocks	How to close
U2	Boost air-temp (pin 79): NTC or ratiometric?	Open	Sheet 3b front-end	One resistance reading, sensor cold — brief task 1.3, closes it outright. **Mitigation: a front-end that works either way 13 Oct 2026:** the Bosch catalogue's T-MAP boost sensors carry a 2.5 kΩ NTC — likely an NTC; the resistance reading still decides it
U3	Controller MPU input — does it exist?	**CLOSED 18 Sep 2026 — No.** The fitted controller is a **DSE4522 MKII**; DSE manual `057-260` 's terminal table runs 1–35 with no speed-pickup terminal	§3	Closed from the manufacturer's own terminal list
U4	Controller J1939 expectations	**Largely answered** — CAN source addresses **234** (engine) / **44** (instrumentation); J1939-75 instrumentation on, alarms off; the controller takes **oil pressure, coolant temperature, coolant level and ENGINE SPEED** from our ECU over CAN; `ECU Data Fail → Shutdown`	Firmware, CAN sheet	Remaining: the exact PGN set. Log the live bus, or read DSE publication 057-004 **13 Oct 2026:** an engine-off trace arrived but holds only UDS diagnostics (0x7E0/0x7E8), no J1939; still needs a running-engine trace on the DSE CAN
U5	Injector part number and drive profile	**Partly answered** — memo 04 gives a family envelope (65–115 V, 12–24 A peak, 8–13 A hold, 40–100 A/ms) sufficient to design the driver	Boost rail *target* voltage, not the driver sheet itself	Photograph the injector body for a Bosch `0445 1xx xxx` or Kirloskar `F6.xxx.xx.x.pr`. **Separately: the calibration data is a different problem — see §9 and memo 04 13 Oct 2026:** KOEL injector number `4H.508.70.0.PR.REV-1` ; the Bosch number on the injector body still needed (`refs/field-evidence-2026-10-13.md`)
U6	Pin 09 `SYNCHRONIZATION GROUND` function	Open	Power/ground plan	OEM pin list; drawn red despite the name
U7	Pins 04/06 — sense only, or load feeds?	Open — no longer blocking	The board now uses them as load feeds (POWER-PATH, 8 Oct 2026), relying on the 10 A loom fuse sizing the wire. Measuring still confirms the OEM's own use	Measure on the live engine, or OEM pin list

#	Unknown	Status	Blocks	How to close
U8	6-pin intake throttle — on the ECU at all?	**Partially answered 18 Sep 2026** — reported **battery-fed**, so the ECU does not supply it	Whether a throttle *driver* is in scope. Supply and control are separate questions and only supply is answered	Still needs brief task 3.11: follow the cable and see whether it terminates at the 94-way connector. A battery-fed actuator can still be ECU-commanded
U9	Catalyst temp sensor — connected where?	Open	Whether an EGT front-end is in scope	Trace on the engine
U10	`SENT` in the colour legend — which signal?	Open	Possible digital sensor front-end	Inspect harness; SENT is SAE J2716
U11	Engine identity	**CLOSED 18 Sep 2026** — rating plate reads **3GK550ETA 4SR1**, app code **GK3.8703**, 26.5 kW at 1500 rpm, type approval `ARAI/MoEF/DGTA/IGES4/KOEL-P25/2825/24`. Memo 01's inferred `3R550ETA 4G1` was **wrong**; the original "GK550" was right	—	Closed by plate
U13	Which controller is fitted	**CLOSED 18 Sep 2026 — DSE4522 MKII AMF (India SP)**, part `4522-001-01`, serial 11021794. Photographed front and rear. **Not a KG640C.** Memo 02 superseded; spec §3 rewritten	—	Closed by photograph
U14	Pre-heat / post-heat: driven by the controller or by the ECU?	**Partially answered 18 Sep 2026** — reported **battery-fed**	Whether a heater *output* is in scope. As with U8, supply is not control	Who switches that battery feed is still open. The DSE4522 config enables both at 50 °C, so the controller is the likely switch — photograph the heater relay and trace its coil
U15	What does `CRS-878` denote in the controller's engine profile?	**NEW 18 Sep 2026**	Possibly bears on U5 — "CRS" plausibly identifies the common-rail system	Ask KOEL, or find DSE's engine-profile list
U17	Which machine are the old ECU (`MD1CS162`, KOEL `T4.435.01.0`) and the 1 Oct CAN trace from?	**NEW 13 Oct 2026** — the trace's ECU reports `KOEL_TC_82.5KVA`, `4R1190ETA4G1` (4-cylinder), not our 25 kVA `3GK550ETA4SR1`	Whether the trace and that ECU describe our engine at all	Ask Jeevan

#	Unknown	Status	Blocks	How to close
U16	**Is the battery-charging alternator's rectifier suppressed (avalanche/clamping), or unsuppressed?**	**NEW 18 Sep 2026. CLOSED 9 Oct 2026 by the owner: design for the suppressed case** (clamped load dump, ISO 7637-2 pulse 5b, about 40 V), which the board already does. The ECU is built from scratch, not matched to this machine's alternator, and modern alternators clamp.	**The input-protection topology.** Suppressed ⇒ ISO 7637-2 pulse 5b, ~40 V clamped, and the fix is a TVS standoff above the dump level — a part number. Unsuppressed ⇒ pulse 5a at 65–87 V unclamped, tens of joules that nothing downstream can ride out, and a different circuit	The alternator's rectifier part number (look for a suppression/avalanche bin), or a scope on the battery terminal during a live disconnect under load. The scope test needs the set running — a supervised activity, not a solo checklist item **Photo 9 Oct 2026** (`refs/charging-alternator-2026-10-09.jpg`): a compact belt-driven unit with the regulator and rectifier inside a black rear cover, a stud B+ terminal and a one-wire plug (green wire). Only production stamps are visible (`G90 518 13`, `30.11.24 21:07`, `09.11.24`, so it was built Nov 2024); no catalogue match. Still needed: the maker's label or part number, usually on the rear cover or a sticker
U12	**Does the 70 A ignition relay `-13RB1` remove power from the ECU and fuel metering unit, or only signal the ECU?**	**CLOSED 18 Sep 2026 — it only signals.** Reported from the machine. The relay does not remove power; a hung ECU keeps the fuel metering unit energised	**§3's safety gate, resolved against us.** The trip module is mandatory, and §3 item 5 now requires its contacts to break the metering unit's own battery feed rather than signal the same relay chain	Closed. Task 1.4's normally-closed check is still wanted as confirmation, but nothing depends on it — the unsafe reading is already adopted

~~**U12 now stands alone as the load-bearing unknown.**~~ **U12 closed 18 Sep 2026, and the answer was the bad one.** U13 cost this project its safety argument: the fitted DSE4522 takes engine speed from our ECU over CAN, so the controller's overspeed trip is not independent of the thing it is meant to protect against. What independence remained was the generator over-frequency shutdown at 56 Hz — and every protective action the controller can take terminates in the same fuel relay on terminal 3.

That relay signals. It does not remove power. So the second line of defence can detect a runaway and cannot stop one. The two findings compound: the controller either cannot see the fault (overspeed, read from our CAN) or cannot act on it (over-frequency, acting through a signalling relay).

Nothing on this machine currently interrupts fuelling without the cooperation of the ECU being protected against. The trip module in §3 item 3 is what closes that, and §3 item 5 now specifies where its contacts have to sit — in series with the metering unit's own battery feed, not in the GCU relay chain that just failed this test.

Worth stating plainly, because it is the pattern of this whole phase: **four of the five unknowns closed by field evidence closed against what this project had reasoned.** U11 (engine identity), U13 (which controller), U1 (connector — this one closed *for* us), and now U12. Desk inference has a measured track record here, and it is poor.

U1, U3, U11 and U13 all closed on 18 September from photographs and one manufacturer's manual. Three of the four closed *against* an inference this project had made, which is the honest measure of how much desk research was carrying.

U3 and U11 were the other two on that list and are now answered, both by inference from primary documents rather than by confirmation. Neither answer is fragile — but both are the kind of finding that a single photograph at the machine would settle permanently, which is why they stay on the register rather than leaving it.

U5 has moved off the critical list for a reason worth stating plainly: memo 04's family envelope means the **injector driver can be designed now, without the part number**. What the part number buys is a boost-rail target instead of a range. The thing that genuinely gates running this engine is not on this register at all — it is the injector calibration data, recorded in §9 and memo 04.

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9. Out of scope

- Injection calibration: timing maps, quantity maps, rail pressure setpoint schedules. The hardware will be capable; the maps are a dyno programme, not a PCB deliverable.
- Emissions certification. **But read this before planning deployment** — research memo 06 found, in primary sources, that the regulatory position is worse than "we simply won't certify it." CPCB's RECD System & Procedure states that ECU characteristics governing injection timing, air-mass metering and emission-reduction strategy must not be altered even under CPCB's own sanctioned retrofit scheme, and excludes changes limited to engine control from that scheme entirely. General Condition 3 of GSR 804(E) states that no person shall use a genset lacking a valid Type Approval and CoP certificate. No approval pathway for a third-party ECU replacement was found.

This does not block the engineering: designing the board, building it, and bench-testing it against simulated signals are unaffected. It bears on **putting it into service on a genset that is in use**, and it compounds the calibration risk in memo 04 — this engine meets its limits partly through calibration, so a replacement ECU can move it out of compliance without any hardware changing at all.

This is the project owners' decision to take to qualified counsel or to CPCB/ARAI directly. It is recorded here because it is material and was not previously examined, not to make the decision for them. - The panel controller. We talk to the **DSE4522 MKII AMF**; we do not replace it. (Earlier revisions of this line said KG640C, which is what the OEM wiring diagram shows and not what is fitted — see §3.) - Panel wiring, ATS, alternator metering.

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10. Phase sequence

Phase	Output	Status
0	This document	Done
1	Research memos: engine ID, controller, MCU confirmation, injector and connector sourcing	**Done** — nine memos
1.5	**Block-level simulation: topology and component values verified before capture**	**In progress — 17 blocks, all passing**
2	KiCad hierarchical schematic capture	Not started
3	4-layer PCB layout, DRC, fab outputs	Not started
4	Firmware skeleton: HAL, J1939, crank sequencing, fault manager	Not started

Phase 1.5 was added on 18 Sep 2026 to name work that was already happening. The simulated blocks are neither research memos nor schematic capture, and leaving them unnamed meant the roadmap did not describe the most productive activity on the project — the one that caught a divider that would have destroyed an ADC pin, a discrete input that was impossible as specified, and a decoupling network thirty times over its impedance target. It also gives the two gaps the circuit review found — load dump, and injector turn-off energy — a phase to belong to instead of floating as review findings.

Phase 2's blocked sheets are now fewer: **U1 is closed** (Bosch `1 928 405 192` / `194`, code C), so the connector and adapter-harness sheets are unblocked. U2 blocks one sensor front-end, U8 blocks whether a throttle driver exists at all, and U5 sets the boost rail's target voltage but not the driver itself. Power, CAN, MCU support, speed inputs, the relay outputs and the analog front-ends are all fully specified above.